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Physics-informed deep learning quantifies propagated uncertainty in seismic structure and hypocenter determination

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Subsurface seismic velocity structure is essential for earthquake source studies, including hypocenter determination. Conventional hypocenter determination methods ignore the inherent uncertainty in seismic velocity structure models, and the impact of this oversight has not been thoroughly investigated. Here, we address this issue by employing a physics-informed deep learning (PIDL) approach that quantifies uncertainty in two-dimensional seismic velocity structure modeling and its propagation to hypocenter determination by introducing neural network ensembles trained on active seismic survey data, earthquake observation data, and the physical equation of wavefront movement. An analysis of an earthquake in southwest Japan using our method revealed that accounting for such uncertainty propagation significantly reduced the bias and uncertainty underestimation in the hypocenter determination, enabling quantitative evaluation of the focal depth relative to the plate boundary. Our results highlight the potential of PIDL for various geophysical inverse problems, such as investigating earthquake source parameters, which inherently suffer from uncertainty propagation.

In seismogenic zones, realistic subsurface seismic velocity structure models provide fundamental information to understand seismic phenomena. In addition to geophysical interpretation of the structures, they also serve as input data for subsequent analyses such as earthquake source inversions (ESIs), including hypocenter location[1–3], moment tensor[4–6] and finite fault[7–9] of ordinary and slow earthquakes. The estimation of such seismic velocity structures has been primarily based on local earthquake tomography using natural earthquakes[10–12]. In addition, high-resolution seismic velocity structure models, mainly in two-dimensional (2D) profiles, for shallow portion of subduction zones that host megathrust earthquakes have been obtained through travel-time inversion, such as first-arrival travel time tomography[13–15], and full waveform inversion[16–18] of active-source seismic data. Efforts are now emerging to incorporate this information and create three-dimensional (3D) velocity structure models[19–23]. Seismic velocity models enhanced by active-source exploration data are expected to lay the foundation for accurate monitoring of faulting activities through ESIs.

However, careful consideration is required for the dependency between inherent uncertainties in seismic velocity structure estimations and ESIs. The estimation of the seismic velocity structure is a highly nonlinear inverse problem and is conducted using data limited to the Earth's surface. These features lead to large uncertainty in the estimation results at certain depth even in a well-configured 2D estimation using active-source seismic data[15]. Due to this uncertainty, the possibility arises for proposing multiple different velocity structure models (Fig. 1AB). A typical ESI is conducted by selecting a single-velocity structure model such as the average one, often with further model simplification by individual researchers and institutions. The differences in models assumed by various groups can cause significant variations in estimated source models for the same earthquake, making the reliability of scientific findings unclear[24] (Fig. 1C). A core of this issue lies in the fact that the propagation of uncertainties from the seismic velocity structure model to ESI is ignored by selecting a single model in advance, resulting in bias and underestimation of uncertainty in ESI[25–28]. For accurate and reliable ESI, in addition to quantifying the uncertainties in ESI itself[29–33], quantification of uncertainty in the velocity structure estimation and its propagation to ESI must be properly considered.

Although several efforts have been made to incorporate such uncertainty propagation in ESIs, these were limited to the application of simplified velocity structure models with roughly assumed uncertainty[26,27,34,35] and other methods based on approximated non-diagonal data-error covariance matrices[25,36–38]. The impact of propagation of genuine uncertainty inherent in seismic velocity structure estimation to ESIs has not yet

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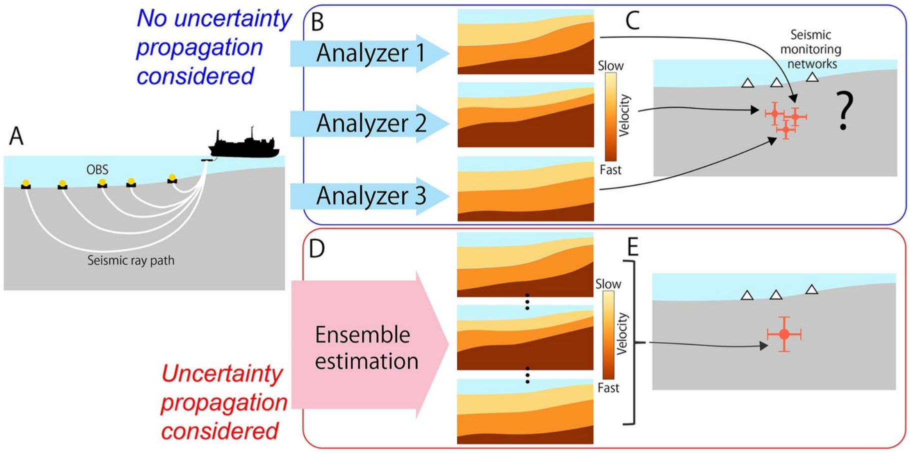

Figure 1. A schematic illustration of the comparison between full UQ-based ESE and conventional (without full UQ) approaches. (**A**) Example of seismic data acquisition for the estimation of the velocity structure using a marine airgun-OBS seismic refraction survey. The dependence of the seismic ray path on the velocity structure leads to strong nonlinearity in the estimation problem. (**B**) Schematic illustration for the seismic velocity structure estimation in conventional non-full UQ-based methods, in which the seismic velocity structure models estimated by different individual analyzers using the same or similar seismic datasets may significantly differ owing to the inherent uncertainty in tomographic analyses. The models may be further simplified by the analyzers. (**C**) Schematic illustration of hypocenter determination using the arrival time data from ocean-bottom seismic monitoring networks (white triangles) as an example of ESE using the obtained seismic velocity structure models. The orange cross marks schematically correspond to the confidence intervals of hypocenter determination. It indicates that different seismic velocity structure models may provide different hypocenter estimates that are biased by the underestimation of uncertainty (in such cases, the extent to which the scientific findings can be trusted is unclear). (**D**) Schematic illustration of the seismic velocity structure estimation in the proposed method, in which the uncertainty in velocity structure estimation was quantified by estimating the model as an ensemble. (**E**) Schematic illustration of hypocenter determination of the proposed method, in which, by incorporating such a velocity structure ensemble as an input, we can accurately account for uncertainty propagation and eliminate the negative effects of unreasonable model selection and simplification.

been well studied due to technical reasons. Existing uncertainty quantification (UQ) methods for seismic velocity structure modeling adopted ensemble-based approaches such as simple Monte Carlo analyses[15,39] and Bayesian estimation more recently[40–43] to cope with inherently strong nonlinearities. Difficulty in handling high-dimensional parameter spaces in Bayesian estimation in these existing methods often limits the UQ of seismic velocity structure models to lower-dimensional representations[40–42]. If realistic samples are obtained, the uncertainty propagation can be evaluated by inputting the ensemble to ESI based on the Bayesian multi-model approach[27,28,35]. However, samples that does not resemble the real velocity structure accurate, as obtained in these methods, make the evaluation of the uncertainty propagation difficult.

The recent remarkable progress in deep learning has led to the emergence of alternative approaches for solving partial differential equations (PDEs) and PDE-based inverse problems, represented by physics-informed neural networks (PINNs)[44]. When analyzing observational and experimental data, PINN incorporates information from the physical laws described by the PDE. PINN is particularly advantageous in solving ill-posed inverse problems owing to its inherent regularity[45], and has already been applied to several geophysical forward[46–54] and inverse problems[55–58] including seismic tomography[59,60]. The PINN-based seismic tomography has recently been extended to ensemble-based estimation[61]. In contrast to conventional grid-based approaches, this method yields samples that more closely resemble real velocity structures represented by an infinite-dimensional function, likely leading to accurate quantification of the propagated uncertainty in seismic velocity structure modeling to ESI (Fig. 1d,e).

Here, we conducted quantification of uncertainty inherent in seismic velocity structure estimation and its propagation to hypocenter determination, the simplest form of ESI, in a real-world problem for the first time using emerging PINN techniques. The target event is the Mw 5.9 earthquake that occurred in 2016 in the Nankai Trough region of southwest Japan. The experts were initially unsure of whether the earthquake was a plate-boundary earthquake[62,63]. First, we used the first-arrival travel time data obtained from refraction surveys conducted near the hypocenter to conduct UQ for the 2D P-wave velocity structure on the survey line to obtain an ensemble velocity-structure model. We then conducted the UQ for hypocenter determination considering the uncertainty propagation from the estimation of the velocity structure using this ensemble. Moreover, we applied the ensemble to the UQ of the estimation of fault location using seismic reflection survey data collected in the same area. Subsequently, we investigated the fault plane wherein the 2016 earthquake occurred from a statistical viewpoint under properly quantified uncertainty. Through these analyses, we found that accounting for such uncertainty propagation from seismic velocity structure estimation significantly reduces the bias and underestimation of uncertainty in the estimated hypocenter location, enabling quantitative evaluation of the focal depth relative to the plate boundary.

Ensemble-based estimation of the P-wave velocity structure

The 2016 Mw 5.9 earthquake occurred off the southeastern coast of Mie Prefecture[62–64] in the central part of the Nankai Trough region in southwest Japan, which is known for historical megathrust earthquakes[65]. This was the largest earthquake in the region since the 1944 Tonankai earthquake (Mw = 8.0) (Fig. 2a). In the vicinity of the hypocenter, the seismic refraction data acquired using a tuned airgun array and ocean bottom seismometers (OBS) are available along the line KI03. We estimated the P-wave velocity structure and its uncertainty using a PINN-based ensemble estimation method[61] based on 14,146 first-arrival travel times that were manually picked from the refraction data. This method represents the seismic velocity structure and travel time function by using two different type of neural networks (NNs), respectively, instead of grids or meshes. The travel time NN is to be optimized for given velocity structure through the PINN framework by minimizing the residual of the Eikonal equation, which can simulate wavefront movement and determine travel time[46,47]; the evaluation the residual was conducted in a straightforward manner with the help of automatic differentiation of the NN outputs[44]. The velocity structure NN gives the velocity structure to this PINN-based training of the travel time NN. We applied the feed-forward fully connected NNs with five hidden layers with the Swish activation function[66] to both the NNs of velocity and travel time, wherein 32 and 96 neurons per layer were used, respectively. Trainable Fourier feature embeddings[67] were applied to both NNs to enhance the representation power for high frequency components. The NNs were basically initialized based on He's initialization method[68]. We generated an ensemble of velocity structure NNs representing the posterior probability for the travel time data formulated by Bayes' theorem, namely, the stochastic property of the estimation uncertainty. This was achieved through the combined use of PINN-based travel time calculations and a particle-based variational inference (ParVI)[69], which is known to enable more efficient Bayesian estimation than Bayesian sampling method (e.g. Hamiltonian Monte Carlo method[70]) (Fig. 2b). ParVI was performed in the function space inferred by NNs (i.e., function-space ParVI[71]), not in the space of the weight parameters of NNs. As shown in[61], function-space ParVI (fParVI) provides more accurate results than ordinary ParVI when applied to PINN-based Bayesian seismic tomography[72]. We introduced 256 particles, i.e., velocity structure instances, in fParVI to represent the stochastic property of the uncertainty. Mini batch training was performed, i.e., 2,048 data randomly sampled out of 14,146 first-arrival travel times were used in each iteration of fParVI. See Methods for details of improvement of the learning efficiency of high-frequency components[73,74] and the convergence of ParVI[75,76], and the optimizers[77,78], the initialization of NNs, and other technical details.

Using the 256 NNs of velocity obtained through ensemble estimation (Fig. 3a), we obtained the mean model (Fig. 3b) and standard deviation (Fig. 3c) of the predicted velocity models. The obtained mean seismic velocity model clearly show the north-dipping surface of the subducting oceanic plate and low-velocity areas corresponding to an accretionary prism and forearc basin, without the introduction of any prior information. The velocity structure in the low uncertainty region of the estimated mean model, including these structural features, are generally in good agreement with the previous tomographic model estimated deterministically for the same first-arrival travel time data[62,64]. The standard deviation of the ensemble, namely, the uncertainty of

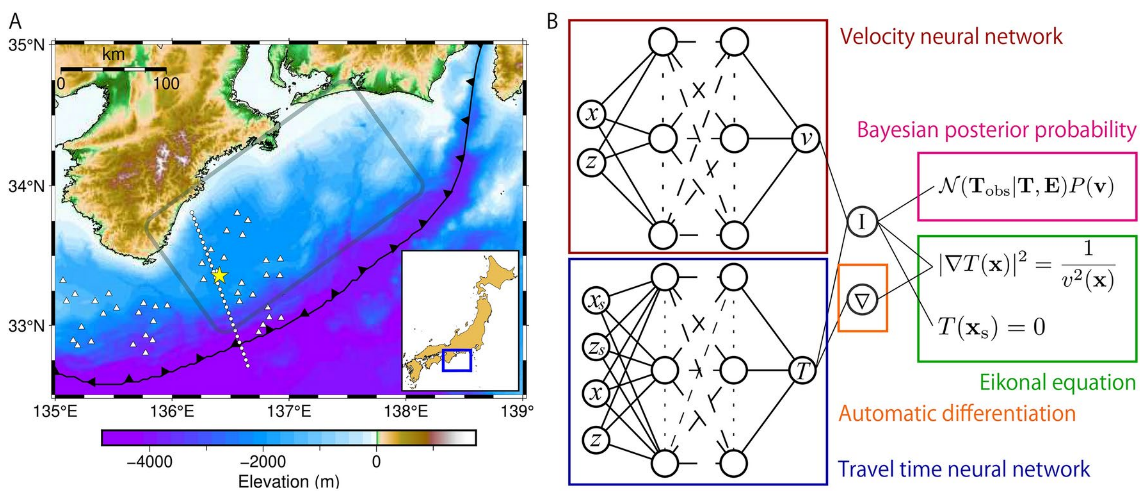

Figure 2. (**A**) A map of the study area. The yellow star represents the epicenter of the 2016 Mw 5.9 earthquake off the southeastern coast of Mie Prefecture estimated in this study. The circles and triangles represent the locations of OBSs, which were installed in the survey line KI03 and DONET observation nodes, respectively. The gray rectangle represents the approximate focal region of the 1944 Tonankai earthquake. The map was created using PyGMT[101], a Python wrapper for GMT6[102]. (**B**) A schematic illustration for the NNs of velocity and travel time trained by both the Eikonal equation based on the PINN framework using automatic differentiation and Bayesian posterior probability based on travel time data. See Materials and Methods for the definitions of the mathematical expressions.

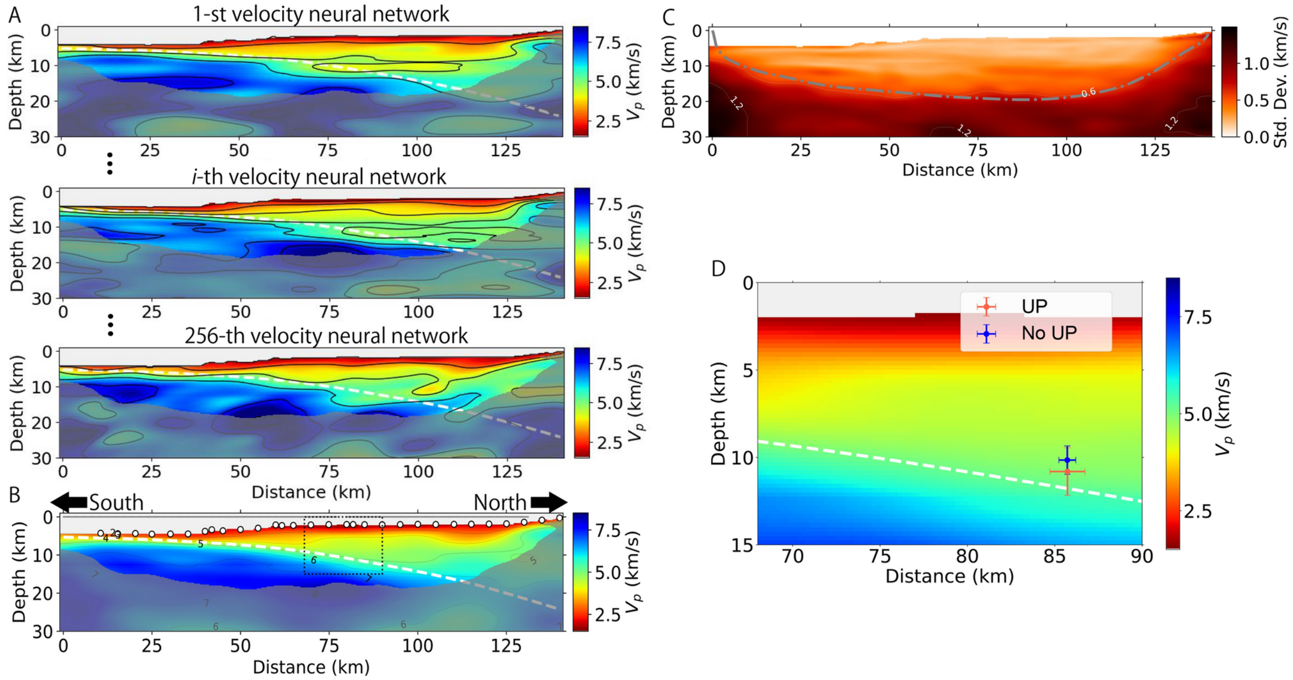

Figure 3. (**A**) Samples from the 256 velocity models along the line KI03 represented by NN ensemble members trained by a combination of PINN and ParVI. In (**A,B**), the gray shaded area represents the region with standard deviations larger than 0.6. In (**A,B,D**), the white dashed line represents the plate boundary model proposed by Nakanishi et al.[21]. (**B**) The mean velocity model calculated based on the estimated ensemble. White circles and gray marks indicate the locations of OBSs and air-gun shots, respectively. (**C**) Standard deviation calculated based on the estimated ensemble. The gray dotted-dashed line denotes the bottom of the ray coverage for the mean model. (**D**) Comparison between the determined hypocenter locations with and without full UQ plotted in the region with the dashed rectangle in (**B**). The cross marks correspond to 2-σ intervals.

the seismic velocities, is generally low in the area covered by the ray path of the first arrivals. It shows spatial variations even within the ray coverage area, suggesting an uneven distribution of ray paths.

Quantification of propagated uncertainty in hypocenter determination based on the ensemble seismic velocity structure model

We re-determined the hypocenter of the 2016 earthquake considering the uncertainty propagation from the estimation of the velocity structure using the ensemble estimation results. We used the P-wave first-arrival time data from nine nearby seismometers installed in the Dense Oceanfloor Network system for Earthquakes and Tsunamis (DONET)[79,80] to estimate the source location[64] (Fig. 4). We set the analysis region to match the arrangement of the observation points. We took such observation and modeling geometry following[64] to narrow the model length in the trench-parallel direction, justifying an assumption that the spatial variation of the seismic velocity structure in the direction perpendicular to subduction is small in the analysis domain. Based on this assumption, we extended the 2D velocity structure models obtained in the previous process to out-of-plane direction and created a 3D velocity structure. Prior to the determination of the hypocenter, we trained the 256 PINNs of 3D travel time function separately for each of seismic velocity structure ensemble and input pairs of the source and receiver points sampled from the targeted regions. The training of each PINN follows the previous approaches to develop surrogate travel time calculation models for arbitrary pairs of source and receiver chosen in the region of interest[3,81] (See Methods for the details of the training). Subsequently, ensemble hypocenter determination was conducted with fast travel time calculations using the pre-trained PINNs combined with a ParVI-based Bayesian inversion method[3]. The uncertainty propagation from velocity structure estimation was considered by integrating 256 velocity models into this hypocenter determination method. This integration was achieved using the Bayesian multi-model approach, a framework that incorporates multiple candidate models in Bayesian inversion[82], which has later been applied to ESIs[27,28,35]. See Methods for the setting of the likelihood function[83] and the optimizer[84], the choice of hyperparameters[85–87], and other details. We introduced 128 particles, i.e., hypocenter parameter instances (particles), in ParVI to represent the stochastic property of the uncertainty (Fig. S1). Full batch estimation was performed, i.e., all nine travel time data were used in every iteration of ParVI.

The mean location of the hypocenter obtained considering the uncertainty propagation were at 136.41°E and 33.36°N, and at a depth of 10.81 km. The standard deviations in the horizontal direction along the cross section, vertical direction, and the other horizontal direction were 0.50, 0.68, and 0.38 km, respectively. These

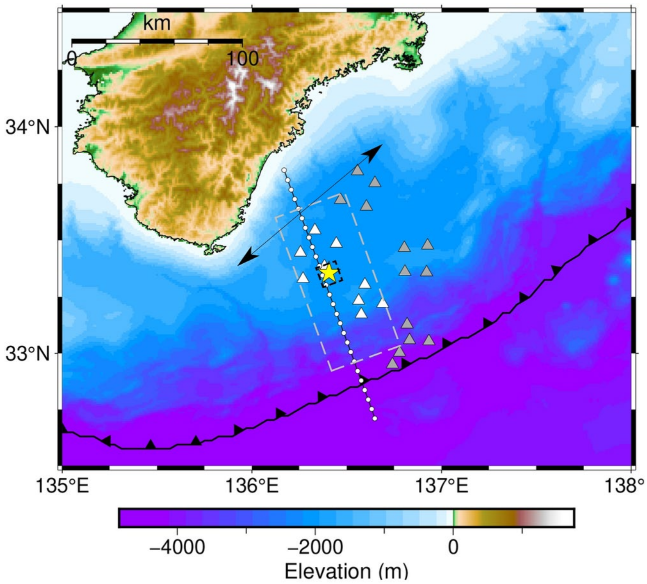

Figure 4. The analysis domains for 3D travel time calculation. The gray dashed rectangle represents the horizontal analysis domain, where collocation points of receivers for PINN training were generated (see "Methods"). The black dashed one represents that for the source modeling, where collocation points of sources were generated. The arrow denotes the direction in which homogeneous seismic velocity distribution is assumed. The triangles and circles denote the locations of the DONET observation nodes that were used (white) and not used (gray) in hypocenter determination and OBSs that were installed in the survey line KI03, respectively. The map was created using PyGMT[101], a Python wrapper for GMT6[102].

values were larger than those estimated without considering uncertainty propagation, wherein the uncertainty of the seismic velocity structure model is not considered (see Methods) (Fig. 3d). The depth obtained without considering uncertainty propagation is reasonably close to that obtained by a previous model with similar analysis conditions[64] (Fig. S2).

Comparison of the locations of the hypocenter and possible coseismic fault under uncertainty

In the target region, two structural interfaces exist near the depth of the plate boundary: a megasplay fault and the top of the oceanic crust. At depth, a plate-boundary earthquake is believed to have ruptured the top surface of the oceanic crust. However, at shallow depths, a plate-boundary earthquake possibly does not rupture the top of the oceanic crust, but ruptures the megasplay fault that branches off from it[88–90]. Here, we consider both structural interfaces as candidates for the "coseismic plate boundary[91]," namely, the fault plane ruptured by the plate-boundary earthquake in this region. We compared our hypocenter with these structural interfaces by considering the uncertainty in their depth related to the seismic velocity structure. Thus, we re-examined whether this earthquake ruptured a coseismic plate boundary and further investigated whether it was located at the megasplay fault, at the top of the oceanic crust, or elsewhere. A seismic section of the two-way travel times (TWTs) is available in the multichannel seismic reflection data for the survey line TK5, which mostly overlaps with line KI03[64]. See Methods for the details of processing of the seismic data[92]. We manually selected reflectors that likely corresponded to a megasplay fault and the top of the oceanic crust (Fig. S3). We obtained seismic sections at depth with uncertainty by converting them to TWT using 256 velocity structure models obtained by using the PINN-based Bayesian tomography (see Methods). Our estimation of the velocity structure utilized airgun-OBS refraction survey data of long offset, which is generally essential for accurate velocity estimations at depths such as 10 km. Consequently, the estimated depths of the plate interface and megasplay fault in the region using this approach are expected to be more reliable than those derived from multichannel seismic reflection data alone, which typically have much shorter offset. The standard deviations of the depths calculated based on the ensemble were 0.3-0.4 km for both reflectors.

Combining all our obtained results enabled the comparative analysis of the locations of the hypocenter and fault under UQ for the first time. The mean location of the megasplay fault agreed well with that of hypocenter depth, and the 2-σ confidence intervals largely overlapped between the hypocenter depth and two fault locations (Fig. 5a). When an earthquake hypocenter is located at either of the two candidate faults or between them, we assumed that the event occurred at the coseismic plate boundary. By leveraging the estimated stochastic properties of the uncertainties, we calculated the probability for this case using a one-dimensional normal distribution fitted to the ensemble assuming a finite fault thickness[93] (see Fig. S4 and Methods). The probability was calculated as 35%. We conducted the same analysis for the case without considering the uncertainty propagation from the seismic velocity structure model. In this case, the candidate fault locations were represented by single lines (Fig. 5b). As previously indicated, the hypocenter estimate is biased toward being shallower with the underestimation

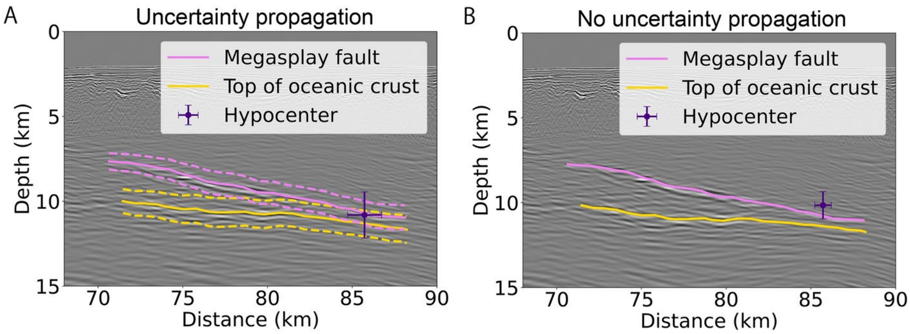

Figure 5. A comparison of the estimated hypocenter and fault locations at depth in the region indicated by the dashed rectangle in Fig. 3b. (**A**) Those obtained using the full UQ-based method, considering the uncertainty in the seismic velocity structure model. The dashed lines and crosses correspond to the 2-σ intervals of the fault locations and hypocenter, respectively. The background image is the seismic section at depth converted from TWT using the mean velocity model. (**B**) The results obtained using an ordinary non-full UQ-based method without considering the uncertainty in the seismic velocity structure model. In this case, the two structural interfaces are denoted by single lines.

of uncertainty. The line of the megasplay fault barely overlaps the 2-σ interval of the hypocenter, whereas that of the top of the oceanic crust is distant from the interval. In this case, the corresponding probability is only 8%.

Discussion

In the proposed approach accounting for the uncertainty propagation from the seismic velocity structure model to hypocenter determination, the probability of the occurrence of an event at the coseismic plate boundary was 35%. This is significantly higher than 8%, which is the probability for the case without considering the uncertainty propagation. Previously, the determination of whether the earthquake was at a coseismic plate boundary was based on the latter type of results, in which the uncertainty in the seismic velocity model is not considered. This latter approach requires an expert's evaluation to consider implicit uncertainties that were not quantitatively considered in the analysis; for example, uncertainty of the velocity structure estimation, effect of the simplification of the velocity structure, and uncertainty of the fault location. In contrast, the proposed method considering the uncertainty propagation allows us to draw insights solely based on the quantified uncertainty in the analysis results, indicating the potential to achieve a more quantitative and objective characterization of various fault activities in seismogenic zones. Future wider applications to other events, including aftershocks of the earthquake targeted in this study and other earthquakes, would confirm such potential of the proposed method.

In the results of the proposed approach, the mean depth of the hypocenters and the two fault locations were similar; thus, the shapes of the probability distributions were similar. Therefore, we expect that improvements in the accuracy and precision of hypocenter determination, fault location estimation, and velocity structure estimation will further increase the probability. For this purpose, the use of the first-arrival time data from other DONET stations distributed over a wider region will provide additional constraints. Since 2019, the Japan Meteorological Agency is expected to issue "Nankai Trough Earthquake Extra Information" to alert the public when the possibility of the occurrence of a megaquake in the Nankai Trough is assessed to be higher than usual[94]. When an earthquake of a certain magnitude occurred in the vicinity of the assumed focal region, whether or not the earthquake occurred at the coseismic plate boundary was the basis for additional information. DONET provides real-time seismic observations at the ocean bottom of the Nankai Trough region as dense arrays. Additionally, rich survey data on the subsurface structures in this region have been compiled in the last two decades[95]. Combined with such abundant data, the proposed method considering the uncertainty propagation is expected to provide quantitative and objective information that is highly useful for both the scientific community and public.

Additionally, we calculated the probabilities of the occurrence of an earthquake at the megasplay fault and the top of the oceanic crust. The results were 5 and 4%, respectively. These low probabilities were largely attributed to the large uncertainty in the estimated hypocenter depth for the thin fault plane. These probabilities also suggest the difficultly in identifying the part of the coseimic plate boundary at which the event occurred based on depth estimates for the dataset we used. Whether or not a large earthquake ruptures a megasplay fault is an important question as seafloor deformation owing to the rupture of megasplay faults may cause larger tsunamis[96]. Although the observation data of the 1944 Tonankai earthquake did not possess sufficient resolution to determine which fault was ruptured, the tsunami inversion result[89] and some interpretations of nearby subsurface structural information[88,90] provide supporting evidence of the slip along the megasplay fault. However, no conclusions have been reached. Further improvements in the precision and accuracy of the involved estimations may provide key information to help conclude these arguments.

We conducted first arrival tomography using active seismic data to obtain seismic velocity models with which we subsequently determined the hypocenter location of a single earthquake. Another possible approach to obtain seismic velocity models in a similar spatial scale is local earthquake tomography, which simultaneously estimates seismic velocity structures and hypocenter locations of natural earthquakes occurred locally[11,12,41]. First arrival tomography and local earthquake tomography generally provide significantly higher resolution for shallow and deep structures, respectively. Well-resolved shallow structure provided by first arrival tomography is essential for accurate hypocenter determination of a relatively shallow event like our target. Recent studies integrated data from natural earthquakes and active seismic survey in tomography analyses[23]. Such efforts combining both methods to obtain seismic velocity models are necessary when both relatively deep and shallow events are targeted.

We here emphasis two major impacts of introducing NN-based techniques that enabled and hypocenter determination considering uncertainty propagation from the models of seismic velocity structures. Firstly, NNs can represent seismic velocity structures in an infinite-dimensional space. This representation agrees better with the nature of the real subsurface structure than with the low-dimensional representation often incorporated in conventional ensemble-based velocity structure estimation methods with grid- or mesh-based parameterization[40–42]. Such a natural representation of the seismic velocity structure is essential for accurate evaluation of uncertainty propagation. This is because accurate and stable subsequent calculations of wavefront movement for the individual velocity models, which are the core of the Bayesian multi-model hypocenter determination, could not be performed if an unnatural low-dimensional representation were adopted. Introducing NN representation of the physical properties into UQ may have the potential for resolving inverse problems involving significant and complex uncertainty propagation. The second aspect is that the determination of the hypocenter based on the Bayesian multi-model framework required only approximately 11 min of computation despite numerous calculations of travel times required: Our Bayesian multi-model hypocenter determination considering propagation of uncertainty in the seismic velocity structure model requires to evaluate travel time for each of 256 velocity models in a single step of optimization process. In addition, the total number of steps was 500, and 128 hypocenter ensembles (particles) were optimized together. Such fast determination was achieved despite so many travel time calculations required because our approach leveraged the pre-trained PINNs, which only need forward propagation calculations of NNs to obtain travel times. The required computation time for the same analysis based on conventional numerical methods such as the fast marching method[97] is at least two orders of magnitude higher than that in our approach as a new numerical calculation is necessary for each instance of the source points and seismic velocity structures. Thus, PINN enabled the fast calculation of the hypocenter determination considering the uncertainty propagation; this is particularly beneficial when quick dissemination of information regarding earthquakes to the public is essential, as in the case of Nankai Trough. These benefits of deep learning approaches are striking examples of how scientific machine learning[98] methods, such as PINN, can lead to new scientific breakthroughs.

A major limitation of our present approach, which is based on the 2D velocity structure estimation, is that the impact of assuming material homogeneity in the trench-parallel direction on the result of the hypocenter determination is not investigated, despite the narrow modeling region taken in that direction. Wider application would definitely require full 3D modeling of the velocity structure. In addition, expanding the target region to incorporate earthquake arrival time data from a wider region, in which the effect of material heterogeneity in the trench-parallel direction cannot be neglected[99], would contribute to higher precision of hypocenter determinations. Using both offshore and onshore data in a wider region would also enable first-motion polarization analysis, facilitating fast determination of earthquake focal mechanisms. Future extension of the proposed method to estimation of 3D models is essential for such further investigations. Analyzing additional events together, such as aftershocks, is also a crucial next step. This approach can improve the precision of hypocenter determinations and enhance our understanding of the sequence of the main shock and aftershocks by narrowing down the solution space of the seismic velocity structure represented by the ensemble model a priori. Another limitation is that some hyperparameters employed in the Bayesian tomography is pre-defined, e.g., the data errors of the travel time picks were defined based on the picker's subjective choice (see Methods). Determining such parameters in a data-driven approach would further increase the objectivity of the estimation results.

The concept of the uncertainty propagation-based approach should be extended from the determination of hypocenter to ESIs in general in the future, including the rupture process and fault slip inversions for coseismic, aseismic, and interseismic faulting activities using seismic waveforms, crustal deformation, and tsunami data. For now, the PINN-based approach in geophysical studies has been most successful in eikonal-based hypocenter determination and tomography, thanks to the simplicity of the formulation and the boundary conditions associated with the equation. Further development of PINN-based approaches to solve PDEs for elastic waves[53] and elastic deformation owing to dislocation sources[50,54] will help the expansion of the uncertainty propagation-based approach to other classes of ESI problems. Such improvements can enhance ESI and various geophysical inverse analyses, which inherently suffer from uncertainty propagation. These advancements are expected to be achieved through the continued development of PINN and relevant scientific machine learning methods as key actors in the "AI for Science" paradigm[100].

Methods

PINN-based ensemble estimation method of P-wave velocity models

We used the first-arrival time data (Fig. S1A) manually picked from the seismic data acquired through the marine Airgun-OBS (ocean bottom seismometer) seismic refraction surveys at KI03 (Fig. 2 and S1), namely, the same dataset used in the estimation of the P-wave velocity model in[62]. The survey was conducted in 2011 by R/V *Kairei* of JAMSTEC. In this survey, 30 OBSs were installed at intervals of approximately 5 km over a distance of approximately 140 km (Fig. S5A); Airgun shots were fired at intervals of approximately 0.2 km at a seawater depth of 10 m, and received by the OBSs. The line KI03 mostly overlaps the line TK5, wherein multi-channel seismic (MCS) reflection survey data acquired in 2001 by the R/V *Kairei* are available, as indicated later. An ensemble of P-wave velocity models in 2D profiles was obtained from first-arrival data using the Bayesian seismic tomography method proposed by[61]. In this method, physics-informed neural networks (PINNs) are combined with a Bayesian inference framework. Two neural networks (NNs) were employed, one of which approximated the P-wave velocity for the input coordinates and the other for the input coordinates of the receiver and source positions (Fig. 3A and S5BC). The NN-based approximation of velocity is defined as follows:

$$\begin{aligned} v(\mathrm{x}) &\simeq f_v(\mathrm{x}, \boldsymbol{\theta}_v) \\ &= \begin{cases} v_{\mathrm{sw}} & (\mathrm{x} \in \Omega_{\mathrm{sw}}), \\ v_0(\mathrm{x}) + f_{v_{\mathrm{ptb}}}(\mathrm{x}, \boldsymbol{\theta}_v) & (\mathrm{x} \in \Omega_{\mathrm{sw}}^{\mathrm{c}}), \end{cases} \end{aligned} \tag{1}$$

where $v(\mathrm{x})$ and $f_v(\mathrm{x}, \boldsymbol{\theta}_v)$ are the true and approximate velocity functions, respectively. x is the input coordinate and $\boldsymbol{\theta}_v$ is the weight parameter of the NN of velocity. v_{sw} is the acoustic speed of seawater. Ω_{sw} and $\Omega_{\mathrm{sw}}^{\mathrm{c}}$ are the analysis domains for seawater and subsurface structures, respectively. $v_0(\mathrm{x})$ denotes the reference velocity set by the user. $f_{v_{\mathrm{ptb}}}(\mathrm{x}, \boldsymbol{\theta}_v)$ is a neural network function to approximate the velocity perturbation component. v_{sw} was set as 1.5 km in this study. $v_0(\mathrm{x})$ was set as a typical one-dimensional (1D; depth-dependent) velocity structure in the target region, determined from a previously proposed P-wave velocity for a nearby survey line[103] (Fig. S6). $f_{v_{\mathrm{ptb}}}(\mathrm{x}, \boldsymbol{\theta}_v)$ were parameterized to ensure that the velocity perturbation ranged from -4.5 to 4.5 km by using the hyperbolic tangent function. However, the lower limit of $f_{v_{\mathrm{ptb}}}(\mathrm{x}, \boldsymbol{\theta}_v)$ was further modified to ensure that the minimum value of $f_v(\mathrm{x}, \boldsymbol{\theta}_v)$ was 1.45 km/s, which is slightly lower than v_{sw}, to maintain the stability of travel time calculation. Fig. S6 shows $v_0(\mathrm{x})$ and the resulting possible value range for $f_v(\mathrm{x}, \boldsymbol{\theta}_v)$. The bathymetric data along the line KI03 were retrieved from the Data and Sample Research System for Whole Cruise Information (DARWIN), made available by[104]. The definition of the travel time NN is based on[61].

We aimed to obtain a velocity structure ensemble that represents the stochastic property of uncertainty in the velocity structure estimation formulated by the Bayes' theorem, considering the observed travel time vector $\mathrm{T}_{\mathrm{obs}}$, each of whose components correspond to the travel time data at each station. Assuming a normal distribution for the likelihood function, the posterior probability density function (PDF) is defined as:

$$\begin{aligned} P(\mathrm{v}|\mathrm{T}_{\mathrm{obs}}) &\propto P(\mathrm{T}_{\mathrm{obs}}|\mathrm{v})P(\mathrm{v}), \\ &= \mathcal{N}(\mathrm{T}_{\mathrm{obs}}|\mathrm{T}, \mathrm{E})P(\mathrm{v}), \end{aligned} \tag{2}$$

where v and T represent the velocity and travel time as the outputs from NNs for the evaluation-point PDE residuals and observation stations, respectively. $P(\mathrm{v})$ represents the prior PDF. E represents the data covariance matrix. The ensemble of velocity NNs to approximate this posterior PDF was obtained using the stein variational gradient descent (SVGD)[69], namely, the best known particle-based variational inference (ParVI) method. The Bayesian sampling from the posterior PDF, e.g., Hamiltonian Monte Carlo (HMC) method[70], is considered to be a standard approach to Bayesian estimation using Neural network (Bayesian NN or BNN). HMC has been applied to Bayesian PINN problems with a relatively small problem size[105,106]. ParVI replaces the Bayesian sampling with an optimization problem applied to a number of instances of the target parameters called particles. In particular, in SVGD, these particles iteratively move toward the posterior distribution, following the gradient of the Kullback-Leibler divergence between the true and modelled posterior PDF, which is obtained from the kernelized Stein discrepancy defined in a reproducing kernel Hilbert space (RKHS). ParVI is known for its high parallelism and efficiency compared with Bayesian sampling methods. As proposed by[61], ParVI was conducted in the function space of the velocity NN, such that the Bayesian estimation for NNs (i.e., BNN: Bayesian Neural Network) could be successfully conducted, even in the high-dimensional parameter space of a NN[71]. During each iteration of the ParVI update in the optimization, the other NN for travel time is trained using the Eikonal equation based on the PINN framework[46,47,51]. The ParVI update requires a gradient of the logarithm of the posterior PDF for the velocity. This was obtained using the adjoint method, assuming that the NN of travel time satisfied the Eikonal equation for the NN of velocity. More details can be found in[61]. We adopted a radial basis function (RBF) kernel to construct RKHS, as in previous studies, wherein its bandwidth was determined using an approach inspired by Scott's rule[75], following a recent study on the acceleration of the SVGD algorithm[76]. We assumed that the data error followed an independent probability distribution and mainly comprised observational errors. Thus, E was set as a diagonal matrix whose diagonal components were set to 0.05, 0.1, or 0.15 s, depending on the accuracy of manual picking of the arrival times. We used a Gaussian process for the prior PDF $P(\mathrm{v})$. $v_0(\mathrm{x})$ was used as the mean function of the prior PDF. The kernel function to determine the covariance of $v(\mathrm{x}_i)$ and $v(\mathrm{x}_j)$ is defined as follows:

$$k_{\mathrm{GP}}(\mathrm{x}_i, \mathrm{x}_j) = \sigma_1^2 \exp\left(-\frac{1}{2\sigma_2^2}|\mathrm{x}_i - \mathrm{x}_j|^2\right), \tag{3}$$

where σ_1 and σ_2 represent the standard deviations of the marginal probability and correlation length scale of P-wave velocity, respectively. To avoid any potential confusion, it is important to distinguish the RBF kernel for the covariance of the prior probability of velocity structure discussed here from the one referenced earlier in the context of RKHS. We further regularized the covariance matrices generated from this kernel function to ensure numerical stability[61]. The hyperparameters included in the RBF kernel were determined through an empirical Bayesian approach using the Widely Applicable Bayesian Information Criterion (WBIC)[86]. The computational cost the of calculation of WBIC for one candidate of the parameter set is equivalent to that of the main analysis. We adopted an empirical Bayesian approach instead of a full Bayesian approach, which estimates the hyperparameters as stochastic variables as well, because the use of the latter approach to determine the hyperparameters in certain forms of prior probability distribution is inappropriate[87] (The details of the two approaches are provided in[85]).

We applied the feed-forward fully connected NNs with five hidden layers to both the NNs of velocity and travel time, wherein 32 and 96 neurons per layer were used, respectively. The Swish activation function[66] was used in each layer except for the output layer, where linear activation was specified. The input coordinates for the NN of velocity were 2D; those for the NN of travel time were 3D as this NN could be further used for the 3D travel time calculation for hypocenter determination, as explained later. In the process of 2D P-wave velocity estimation, only the x and z components were used, and y was fixed as zero. Before passing the input 2D coordinates to the NNs, we incorporated trainable Fourier feature mappings[67], which are a variation of those proposed in[74]. This was done to avoid the bias of the feed-forward fully-connected NN toward low-frequency solutions, a phenomenon known as "spectral bias"[73]. The weight parameters of the velocity NN were initialized using He's method[68]. The velocity structures inferred based on these weight parameters serve as the initial particles for the ParVI optimization. Those of the NN of travel time were trained for the initialized velocity through the PINN framework using a Rectified Adam (RAdam) optimizer[107] for 5,000 epochs.

We used 256 particles for the ParVI optimization of the velocity NN. Thus, we trained 256 pairs of NNs of velocity and travel time in parallel. In determination of hyperparameter σ_1 and σ_2 in Eq. (3) based on the empirical Bayesian approach, a grid search was computationally infeasible. Instead, we first fixed σ_1 to 1.5 km and selected a σ_2 value among 1, 2, 3, 4, 5, and 6 km using WBIC. Although σ_2=2 km provided the largest WBIC, ParVI optimization with this value did not successfully increase the fit to the observational data. Thus, as the second-best choice, we used σ_2=3 km instead. Subsequently, by fixing σ_2 to the chosen value, we set the σ_1 values as 1.0, 1.25, and 1.5 km. We did not consider σ_1 to be larger than 1.5 km as such a value results in an uncertainty that is too large for our prior knowledge. Consequently, σ_1=1.5 km and σ_2=3 km were used. The number of epochs and batch size for the ParVI optimization were 800 and 2,048, respectively. The learning rate was controlled using the Yogi optimizer[77] available in the torch-optimizer package[78]. In the first 600 iterations, the NNs were trained at an initial learning rate of 0.01. We then restarted training at an initial learning rate of 0.001 for 200 epochs. In each iteration, the NN of travel time was optimized for the updated NN of velocity through the PINN framework using the RAdam optimizer for 500 epochs.

The calculation required approximately 100 h using 16 NVIDIA A100 GPUs equipped with Earth Simulator 4, which was made available by JAMSTEC. Sixteen particles were assigned to each GPU. For a wider range of future applications, a reduction in the computational cost is essential. For example, to ensure effectiveness, the number of epochs can be reduced by making better choices for the initial values of the weight parameters in the NNs of velocity and prior probability model.

PINN-based ensemble estimation method of the hypocenter considering the ensemble velocity model

To determine the hypocenter, we used the arrival time data of P-waves processed by[64]. This dataset includes manually picked P-wave arrival times from the vertical component seismograms of broadband seismometers installed at nine stations of the Dense Oceanfloor Network system for Earthquakes and Tsunamis (DONET) located within 30 km of the mainshock source.

Base on[64], we created a 3D velocity volume using 2D velocity structures, assuming that the structure was identical along the strike of the subducting plate. Such a model is often referred to as a "2.5D" model in seismological studies. The 2D structure was oriented in the E40°N direction (Fig. S1). We used this 3D volume with a "2.5D" velocity model to compute the theoretical travel times at each DONET station.

To determine the hypocenter, the uncertainty of the velocity model, which was quantified using the analysis described in the previous section, was considered. This was conducted using the concept of Bayesian multi-model estimation[27,28,35,82]. In an ordinary Bayesian formulation for hypocenter determination without considering the velocity uncertainty, the posterior PDF for hypocenter parameters m is described as follows:

$$P(\mathrm{m}|\mathrm{d}) = \kappa P(\mathrm{d}|\mathrm{m}) P(\mathrm{m}), \tag{4}$$

where $P(\mathrm{m}|\mathrm{d})$, $P(\mathrm{d}|\mathrm{m})$, and $P(\mathrm{m})$ represent the posterior PDF of the hypocenter parameters, likelihood function, and prior PDF of the hypocenter parameters, respectively. $\kappa = 1/P(\mathrm{d})$ is a normalization factor that assumes a constant value as the observation data and model are fixed. In the Bayesian multi-model estimation, we first expand this formulation to a joint posterior PDF with velocity parameters as follows:

$$P(\mathrm{m}, \mathrm{v}|\mathrm{d}) = \kappa P(\mathrm{d}|\mathrm{m}, \mathrm{v}) P(\mathrm{m}|\mathrm{v}) P(\mathrm{v}), \tag{5}$$

where v is the velocity parameter. To obtain the posterior PDF for hypocenter parameters considering the velocity uncertainty, we simply marginalize the equation using the members of the estimated velocity ensemble model or "multi-model" using Monte Carlo integration as follows:

$$P(\mathrm{m}|\mathrm{d}) = \int P(\mathrm{m}, \mathrm{v}|\mathrm{d}) d\mathrm{v}, \\ = \kappa \int P(\mathrm{d}|\mathrm{m}, \mathrm{v}) P(\mathrm{m}|\mathrm{v}) P(\mathrm{v}) d\mathrm{v}. \tag{6}$$

$$\simeq \kappa \frac{1}{N} \sum_{n=1}^{N} P(\mathrm{d}|\mathrm{m}, \mathrm{v}^{(n)}) P(\mathrm{m}|\mathrm{v}^{(n)}), \tag{7}$$

where N is the number of models. Each $P(\mathrm{d}|\mathrm{m}, \mathrm{v}^{(n)})$ can be evaluated through a theoretical calculation of travel time using the n-th velocity model. As discussed in[28], the source inversion methods accounting for prediction uncertainty proposed by[26,34] are simplified versions by applying Gaussian approximation of $P(\mathrm{v})$ and linear approximation of relation between d and m to this formulation.

For the Bayesian estimation of hypocenters, we followed a ParVI-based approach called hypoSVI[3], wherein the travel time between hypocenters and stations was calculated using pre-trained PINNs. We created 128 particles as instances of 3D coordinates of the hypocenter and optimized them using the ParVI framework. Before the optimization procedure, we trained 256 NNs of travel time based on the PINN framework for the 3D Eikonal equation for 256 "2.5D" velocity models, respectively. The 256 NNs of velocity and travel time obtained in the ensemble velocity estimation were directly introduced into this procedure. The former provides velocity values for the PINN-based training of travel time NNs. The latter serves as 3D NNs of travel time by making all 3D input coordinates in effect by allowing non-zero values to be inputted in y, and adopts the weight parameters as the initial guess to speed up the training. Each NN of travel time was trained for random source and receiver inputs. Fig. S1 shows the sampling regions of the source and receiver points in the horizontal direction. The vertical ranges span from depths of 4.6 to 14.6 km for the sources, and from the seafloor to a depth of 40 km for the receivers. Based on some basic investigations, we expect that the travel times inferred by pre-trained PINNs and calculated using the fast marching method agree well with average differences of approximately 20 ms. Assuming that fast marching method is highly accurate, these differences compose major part of the forward modeling errors in the hypocenter determination. Initial values of 128 particles were also randomly chosen from this sampling region. We conducted full-batch training with nine DONET arrival-time datasets by setting the number of epochs to 500. We used the Adam[84] optimizer with an initial learning rate of 0.1. During ParVI optimization, we used these pre-trained travel time NNs to calculate the theoretical travel times, wherein the solution was obtained in a fraction of seconds. The major difference between our method and the original hypoSVI is that we calculated the theoretical travel time 256 times in each iteration for one particle to calculate the Monte Carlo integration in the likelihood function for the Bayesian multi-model estimation, as opposed to only once in hypoSVI. Owing to the fast travel time calculation using the pre-trained PINNs, this analysis required only approximately 11 min using 128 CPU cores (64-core AMD EPYC 7742 × 2 in Earth Simulator 4, made available by JAMSTEC) assigned to each of the 128 particles.

We used an improper flat prior ($p(\alpha) = 1,\ \alpha \in [-\infty, \infty]$) for the hypocenter parameters. We used a likelihood function to eliminate the origin time of the event from the formulation[83]. We assumed that the data errors set in this likelihood function follow an independent and identically distributed (i.i.d.) zero-mean Gaussian distribution. We searched the standard deviations of the data errors from five candidates, 0.5, 0.75, 0.875, 1.0, and 1.25 s, using WBIC without considering the uncertainty in the seismic velocity structure model (see the next section). 0.875 s was chosen as the optimal value for the variance with the maximum WBIC, and was used in the case when also considering the uncertainty in the seismic velocity structure model. The forward modeling errors imposed by PINN-based forward travel time calculations mentioned previously can be a major component of the data errors in the hypocenter determination. This empirical Bayesian estimation accounts for the impact of the forward modeling errors on the precision of hypocenter determinations in part.

Comparison of the results of a case with a single seismic velocity structure model with previous results

Results without considering the uncertainty of the seismic velocity structure model were also obtained for comparison. This is achieved by replacing the velocity structure ensemble in the proposed analysis scheme with a single seismic velocity structure model. In the ensemble estimation of the P-wave velocity, setting the number of ensemble members to one provides a result equivalent to that of the maximum a posteriori estimation[69]. The number of epochs for the ParVI optimization was 700. The initial learning rate was 0.01 and restarting was not conducted. The other parameters were the same as those in the case of the 256 seismic velocity structure models. The obtained single velocity model was then used for ensemble hypocenter determination. All parameters for this analysis were the same as those for the 256 seismic velocity structure models.

Figure S2 shows a comparison of the results of hypocenter determination between the 256 seismic velocity structure models and single model, as well as those estimated by[62] and[64]. Ref.[62] used the 1D velocity structure to calculate the theoretical travel time based on the seismic tomographic results along KI03.[62] used the tomographic result directly based on "2.5D" velocity modeling, which is the approach followed in this study. As our approach for the case with a single seismic velocity structure model is theoretically equivalent to that of[64], the mean depths of hypocenter in these two cases showed the best agreement among the four results.

Processing of seismic sections

A seismic reflection survey at the line TK5 was conducted in 2001 by the R/V *Kairei* of JAMSTEC. The data were collected using a 5.5-km-length hydrophone streamer cable and an airgun-array with a total volume of 127.8 L (7,800 inch^3) fired at intervals of 50 m along the line. The collected data were processed using a fundamental common midpoint (CMP) method (e.g.,[92]) with noise attenuation and multiple reflections. A seismic section of the two-way travel times (TWTs) was obtained using CMP stacking with normal moveout correction and post-stack time migration. Before visualizing and manually selecting the reflectors corresponding to the megasplay fault and the top of the oceanic crust, we applied time-variant dip filters to eliminate the residual multiple reflections interfering with the target reflections (Fig. 5 and S3).

The seismic sections in the TWT can be converted into depth sections using the seismic velocity structure model. We considered the uncertainty of the seismic velocity structure model as the source of uncertainty in the converted depth. We conducted 256 conversions using the velocity structure ensemble and obtained 256 different depth models of the two reflectors, which we considered as the ensemble fault location model.

Statistical analysis of the estimated stochastic property of the depth uncertainties

We calculated the respective probability that the hypocenter is located at each of two candidates of the coseimic plate boundary, the megasplay fault and the top of the oceanic crust. We compared the depth of the hypocenter with those of the two reflectors at the horizontal position of the mean hypocenter. We obtained a 1D normal distribution based on the sample mean and standard deviation of the hypocenter depth. Each member of the ensemble seismic velocity structure model was associated with the different depths of the two reflectors (Fig. S4A). Subsequently, assuming a finite fault thickness, which is set to 100 m[93], we calculated the probabilities for three cases: (A) hypocenter located at either of the two candidate faults or between them; (B) hypocenter located at the megasplay fault; and (C) hypocenter located at the top surface of the oceanic crust. We calculated the target probability by averaging the probabilities of the 256 seismic velocity structure models. Without considering the uncertainty of the seismic velocity structure model, the same calculation was conducted for a single seismic velocity structure model (Fig. S4B). Notably, the probabilities of Cases (B) and (C) were sensitive to the choice of fault thickness.

Data availability

The seismic survey data in the line KI03 and TK5 is available from JAMSTEC (Japan Agency for Marine-Earth Science and Technology) Seismic Survey Database[108] (https://www.jamstec.go.jp/obsmcs_db/e/). The bathymetric data in the Data and Sample Research System for Whole Cruise Information (DARWIN)[104] (https://www.godac.jamstec.go.jp/darwin/en/) were used.

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